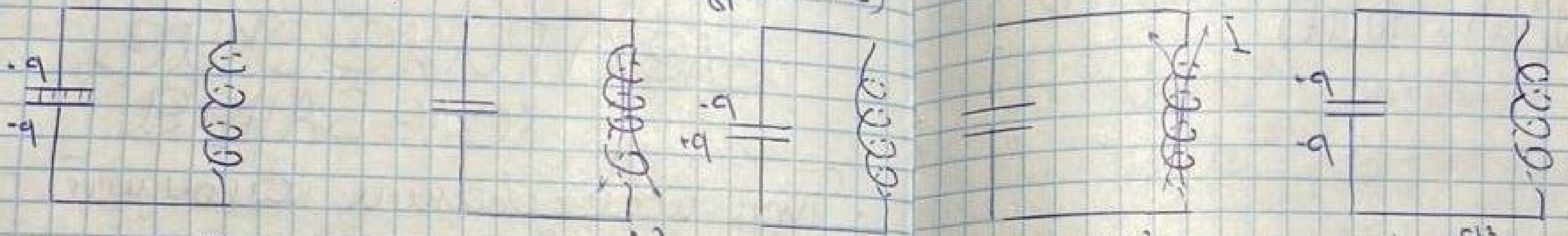
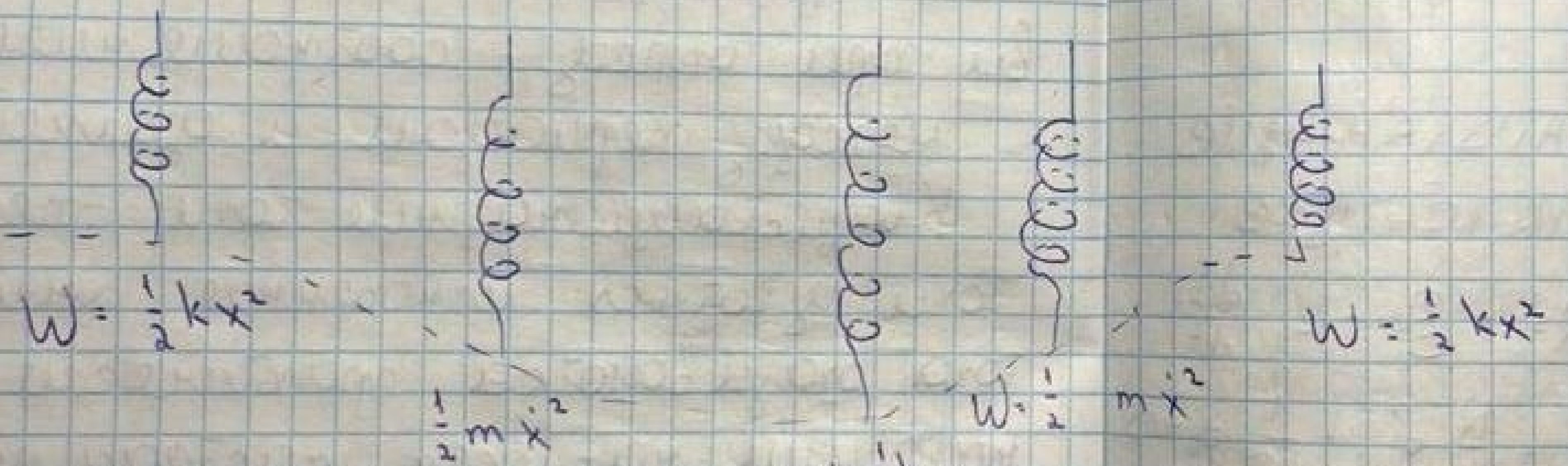
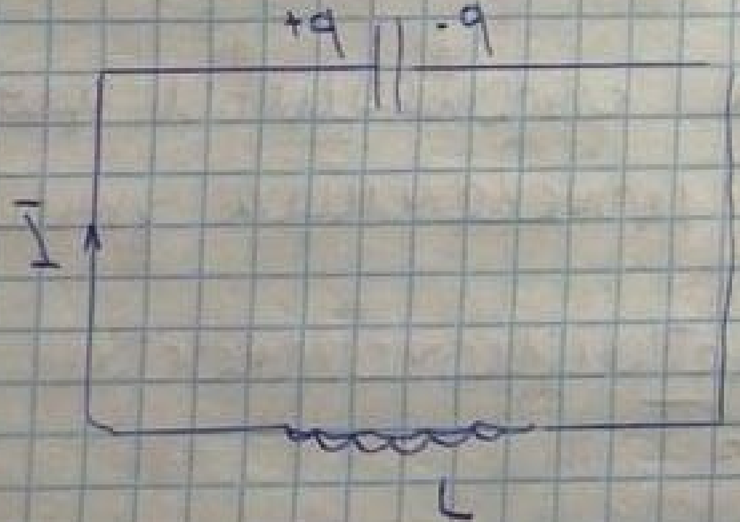
# Свободные колеб. заряда в идеальном контуре (R=0)




$W = \frac{q^2}{2C}$ $\quad W = \frac{L\dot{q}^2}{2}$ $\quad W = \frac{q^2}{2C}$ $\quad W = \frac{L\dot{q}^2}{2}$ $\quad W = \frac{q^2}{2C}$



$W = \frac{1}{2}kx^2$ $\quad \frac{1}{2}m\dot{x}^2$ $\quad W = \frac{1}{2}kx^2$ $\quad W = \frac{1}{2}m\dot{x}^2$ $\quad W = \frac{1}{2}kx^2$




$IR = \varphi_1 - \varphi_2 + \mathcal{E}_{12}$

$R = 0, \ \varphi_1 - \varphi_2 = -q/C$

$\mathcal{E}_{12} = -L\frac{dI}{dt}$

$L\frac{dI}{dt} + q/C = 0$

$\ddot{q} + \frac{q}{LC} = 0, \quad \omega_0 = \frac{1}{\sqrt{LC}}$

$\ddot{q} + \omega_0^2 q = 0$

$q = q_m \cos(\omega_0 t + \alpha)$

$U = \frac{q_m}{C}\cos(\omega_0 t + \alpha) = U_m \cos(\omega_0 t + \alpha)$